

NGUYEN HONG PHUC

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EDUCATION

VinUniversity

Bachelor of Science in Electrical and Computer Engineering

Ha Noi, Vietnam

Oct. 2023 – Present

- **Cumulative GPA: 3.80/4.00 (Rank 1 in ECE Cohort 4)**
- **Honors:** 2 times President's List (AY 23-25), Dean's List (All Semesters), 100% Talent Scholarship (2023–2027).
- **Awards:**
 - * Best Poster Awards, Open Innovation Conference 2025.
 - * Gold Medal – WorldQuant Challenge.
 - * Australia-Vietnam Strategic Technologies Scholarship.
 - * Silver Medal – The University Physics Competition 2024.
 - * Finalist – 2025 Quantum Global Industry Challenge (Connected DMV).
 - * Finalist – IoT Challenge 2025 (FPT Software).

Bac Ninh High School for the Gifted

Physics Major

Bac Ninh, Vietnam

Aug. 2020 – May 2023

- **GPA: 9.6/10**
- Selected Awards: Second Prize National Physics Olympiad (2023), Gold Medal High School Olympiad (2021-2022), 5th Place Worldwide Thomas Jefferson Physics Olympiad (2023).

TECHNICAL SKILLS

Languages: Python, C/C++, Verilog, MATLAB, LaTeX

Frameworks/AI: PyTorch, TensorFlow, Scikit-learn, OpenCV, PennyLane, Qiskit

Tools & Platforms: Docker, Git, Vivado, Altium, PSpice, Arduino, Raspberry Pi, Linux, Teensy

PUBLICATIONS

Journal Papers

- [1] **H.-P. Nguyen**, T. T. Nguyen, S. H. Wanasekara, N. M. Ngo, D. V. Dzung and V.-D. Nguyen, "AquaSC: Task-Oriented Semantic Communication for Efficient and Intelligent Aquaculture IoT Systems," *IEEE Transactions on Emerging Topics in Computational Intelligence* (Submitted 10/2025. **under review**).
- [2] **H.-P. Nguyen***, B. H. Ngo*, M. P. Tran and V.-D. Nguyen, "AquaDeHi: Detached Hierarchical Multi-Task Learning with Learnable Prototypes for Fine-Grained Aquatic Species Recognition," *IEEE Transactions on Multimedia* (Submitted 4/2026. **under review**).
- [3] Q.-H. N. Dam, **P. H. Nguyen**, et al., "From Monitoring to Action: A Comprehensive Survey of Autonomous Decision-Making in Smart Aquaculture Systems," *Computers and Electronics in Agriculture* (Submitted 4/2026. **under review**). Preprint: 10.13140/RG.2.2.24130.95688
- [4] N. M. Ngo, T. T. Nguyen, **H.-P. Nguyen** and V.-D. Nguyen, "Energy-Aware Resource Allocation for Energy Harvesting Powered Wireless Sensor Nodes," *IEEE Communications Letters*, vol. 29, no. 3, pp. 542-546, Mar. 2025.

Conference Papers

- [1] **P. H. Nguyen**, T. T. Nguyen, Q. N. Duong, V.-D. Nguyen, "ChronoSC: Task-Oriented Semantic Communication via Temporal-to-Color Encoding," *IEEE ICCE 2026* (**accepted**).

Others

- [1] **Phuc Nguyen H**, "C-MambaPose: A Physics-Informed Complex Mamba Framework for Cross-Environment WiFi Human Pose Estimation," *Arxiv preprint* (arxiv.org/abs/2606.13700).

WORK EXPERIENCE

Undergraduate Research Assistant

Oct. 2023 – Present

Intelligence, Computing and Communication Lab (ICCL) | VinUni & Trinity College Dublin

- Investigate Open RAN architecture, 5G networks, and Task-Oriented Semantic Communication under Supervisor Nguyen Van Dinh.
- Develop multi-modality AI models for smart aquaculture, implementing Computer Vision algorithms for fish tracking, underwater size estimation, and image segmentation.
- Explore Diffusion Models for enhanced data generation and signal processing applications.

Project Intern (Top 20 Finalist IoT Challenge)

May 2025 – Sep. 2025

FPT Software

Ha Noi, Vietnam

- Deployed Quantized LLMs on Raspberry Pi to create a conversational conservation chatbot.
- Engineered a full-stack solution (Frontend/Backend) for real-time shopper tracking and route optimization via BLE localization.

Research Consultant

Sep. 2024 – May 2025

WorldQuant

Remote

- Developed predictive mathematical models (Alphas) to forecast financial instrument price movements.
- Utilized statistical analysis and historical data to optimize trading signals.

PROJECTS

Deep JSCC for Visible Light Communication | PyTorch, Analog Circuit, Signal Processing

Jan. 2026

- Designed a hardware-in-the-loop Deep Joint Source-Channel Coding (Deep JSCC) system transmitting semantic features via analog light intensity using a custom 8-bit R-2R DAC.
- Implemented an Autoencoder in PyTorch with differentiable channel approximation to learn robust feature representations against non-linear hardware noise.
- Achieved 98% compression (MNIST) and optimized transmission via a custom channel linearization algorithm.
- GitHub: [Code & Demo](#)

Smart Aquaculture IoT System | Multimodal AI, LLM RAG, Hydrophone, Full-stack

In-Progress

- Developed a multi-sensory IoT node utilizing water quality sensors, underwater cameras, and hydrophones for real-time pond health monitoring and server-side processing.
- Engineered a **Multimodal AI** framework fusing visual and acoustic data to optimize feeding schedules, alongside time-series forecasting for water quality trends.
- Integrated an **LLM with Retrieval-Augmented Generation (RAG)** into a Web App to provide context-aware technical advisory services to farmers.

Teensy 4.1 Flight Controller Firmware | C++, Embedded Systems, Control Theory

Jan. 2026

- Developed custom C++ firmware on Teensy 4.1 implementing a 400Hz PID control loop for Quad-X stabilization.
- Integrated Madgwick sensor fusion for precise attitude estimation (MPU6050) and real-time RC signal noise filtering.
- Engineered safety arming logic and optimized Serial plotting (500k baud) for PID tuning and diagnostics.
- GitHub: [Code & Demo](#)

WinCart | Edge AI, Raspberry Pi, Silicon Labs

Completed

- Engineered an AI-powered smart shopping assistant enabling hands-free experiences.
- Implemented Real-Time Positioning via ML on Silicon Labs hardware and a full-stack Edge AI pipeline.

Neural Network From Scratch | C++, C, Mathematics

Completed

- Implemented a Convolutional Neural Network (CNN) in C++ and Feedforward NN in C for MNIST classification without external ML libraries, demonstrating deep understanding of backpropagation and optimization.
- GitHub: [CNN \(C++\)](#) | [NN \(C\)](#)

CERTIFICATES & COURSES

Specializations: Digital Signal Processing (EPFL), Neural Networks and Deep Learning (DeepLearning.AI), Wireless Communications (Yonsei Univ).

Other: Open RAN 101, Quantum101, Introduction to Optimization (VinUni), Introduction to Microelectronics.

LANGUAGES

Vietnamese: Native | **English:** Working Proficiency

FUNDING & GRANT

Student Research Grant 3500\$: Lead research about smart aquaculture